

CTA200 2025 — Assignment 3

May 2025

1 Question 1: The Mandelbrot Set

1.1 Method

For each point $c = x + iy$ in the complex plane, with $-2 < x < 2$ and $-2 < y < 2$, we iterate the map

$$z_{i+1} = z_i^2 + c, \quad z_0 = 0. \quad (1)$$

A point is classified as *diverging* if $|z|^2 = \Re(z)^2 + \Im(z)^2 > 4$ at any iteration step, and *bounded* otherwise (after a maximum of 100 iterations). The iteration function `mandelbrot_iterate` is defined in `mandelbrot.py` and imported into the notebook.

1.2 Results

Figure ?? shows the binary classification: points that diverge are shown in red, and those that remain bounded form the Mandelbrot set shown in blue. The characteristic cardioid body and circular bulbs of the Mandelbrot set are clearly visible.

Figure ?? colours each diverging point by the iteration number at which it first exceeded the escape radius $|z| = 2$. Points that escape quickly (far from the set boundary) appear in lighter colours, while points near the boundary take many iterations to escape and appear darker.

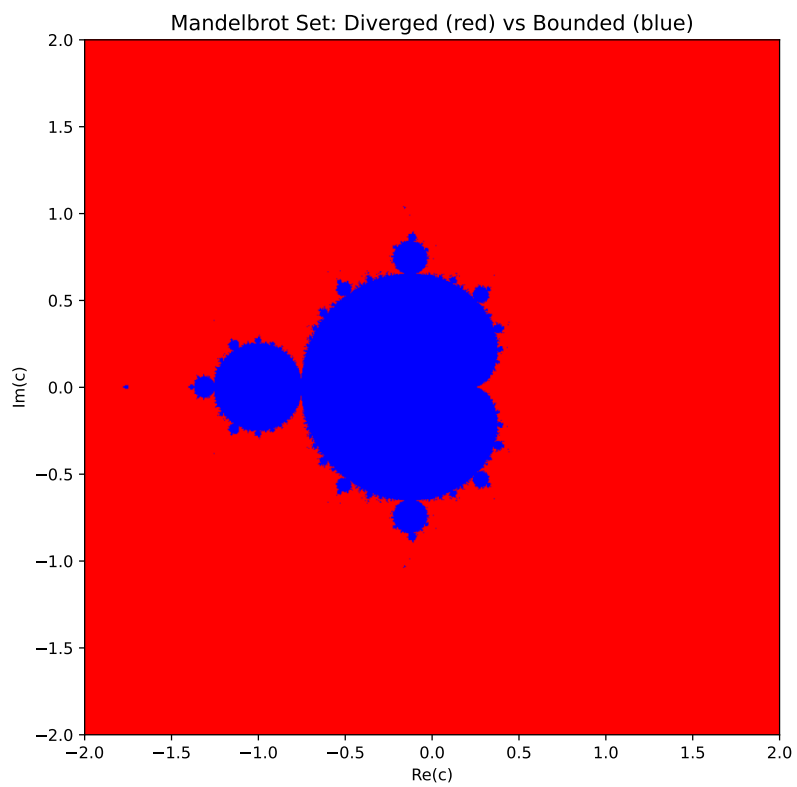


Figure 1: Binary Mandelbrot set image. Red: points that diverge. Blue: bounded points.

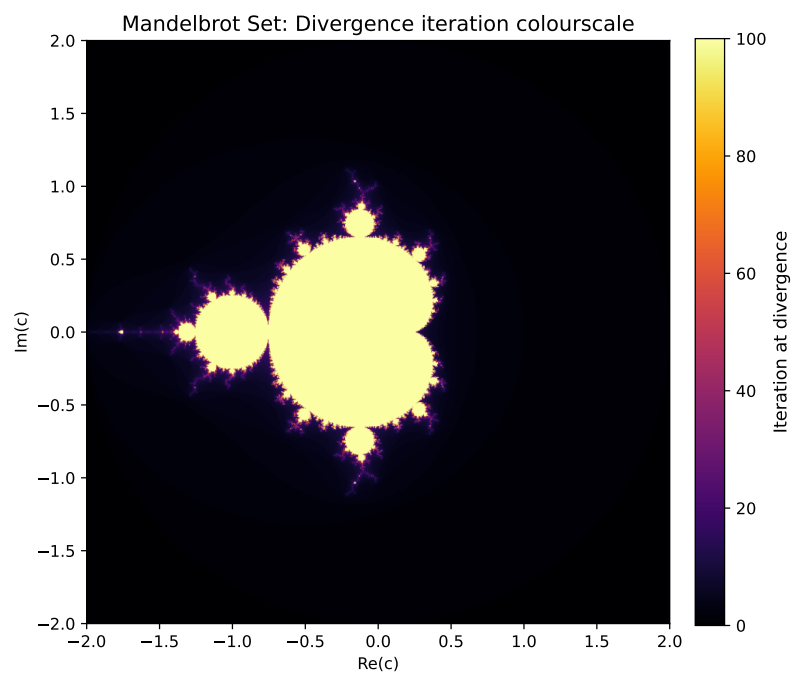


Figure 2: Mandelbrot set coloured by the iteration number at which each point diverges. Bounded points (the set itself) are assigned the maximum iteration count.

2 Question 2: The Lorenz System

2.1 Method

We integrate the Lorenz (1963) system of ODEs,

$$\dot{X} = -\sigma(X - Y), \quad (2)$$

$$\dot{Y} = rX - Y - XZ, \quad (3)$$

$$\dot{Z} = -bZ + XY, \quad (4)$$

using `scipy.integrate.solve_ivp` with the `dense_output=True` option (which gives a continuous interpolant). The Lorenz parameter values are $\sigma = 10$, $r = 28$, $b = 8/3$, and the initial conditions are $\mathbf{W}_0 = (X_0, Y_0, Z_0) = (0, 1, 0)$. Integration runs from $t = 0$ to $t = 60$ (dimensionless units), with tight tolerances (`rtol=1e-10`, `atol=1e-12`) to minimise numerical error. The right-hand side function `lorenz` is defined with a proper docstring in `lorenz.py`.

2.2 Results

Figure ?? reproduces Lorenz’s Figure 1: the variable Y as a function of the iteration index $N = t/\Delta t$ (with $\Delta t = 0.01$). The solution is clearly aperiodic and does not repeat.

Figure ?? reproduces Lorenz’s Figure 2: the phase-plane projection of Z against Y for $t \in [14, 19]$. The trajectory traces the two-lobed structure now known as the Lorenz attractor.

Figure ?? demonstrates sensitivity to initial conditions. A second trajectory is computed with initial conditions $\mathbf{W}'_0 = (0, 1 + 10^{-8}, 0)$, differing from $\mathbf{W}_0$ by only 10^{-8} in the Y component. The Euclidean distance $|\mathbf{W}(t) - \mathbf{W}'(t)|$ between the two trajectories grows exponentially (straight line on the semilog plot) until the solutions saturate at order-unity separation around $t \approx 30$. This is the hallmark of chaos: arbitrarily small errors in initial conditions lead to completely different long-term predictions, making the system practically unpredictable.

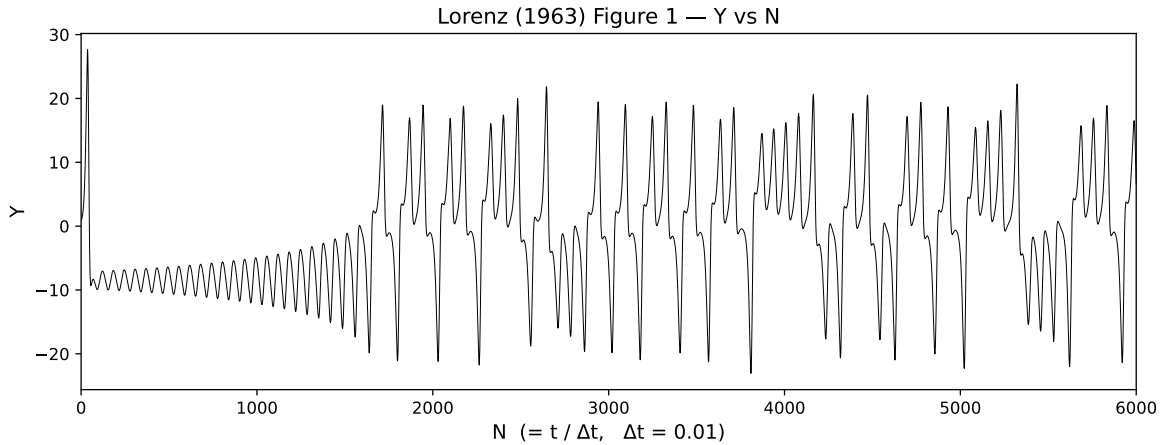


Figure 3: Reproduction of Lorenz (1963) Figure 1. Y as a function of $N = t/\Delta t$, $\Delta t = 0.01$.

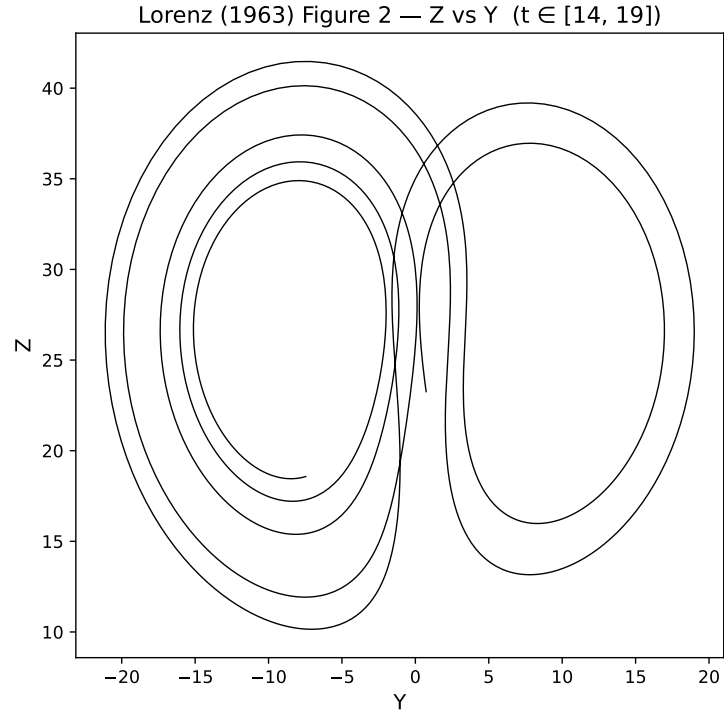


Figure 4: Reproduction of Lorenz (1963) Figure 2. Phase portrait Z vs Y for $t \in [14, 19]$.

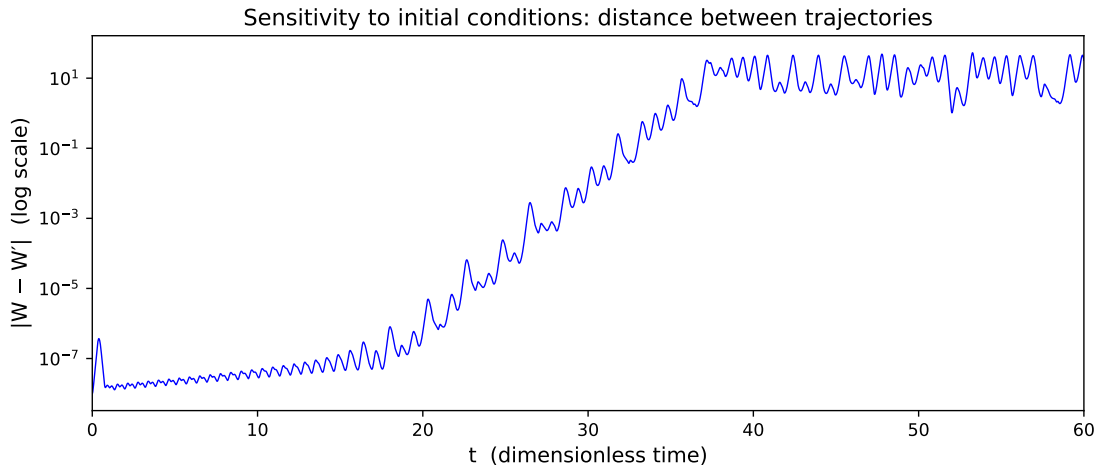


Figure 5: Euclidean distance between two trajectories with initial conditions differing by 10^{-8} in Y , plotted on a semilogarithmic scale. The exponential growth phase (straight line) demonstrates the sensitive dependence on initial conditions characteristic of chaotic systems.